

Airline Flight Price Exploratory Data Analysis

Business Objective

Airline ticket pricing is influenced by a combination of operational characteristics, market demand, and service offerings. Understanding these factors is essential for airlines seeking to optimize pricing strategies and for travelers aiming to make cost-effective booking decisions. This project conducts a comprehensive **Exploratory Data Analysis (EDA)** on an airline flight dataset containing approximately **300,000 flight records** to identify, quantify, and interpret the major factors associated with airfare variation.

Data Quality Assessment

Prior to analysis, the dataset underwent a systematic quality assessment to ensure the reliability of the insights generated.

Data Preparation

- Removed unnecessary columns.
- Verified data types and converted variables where required.
- Examined summary statistics to understand the distribution of numerical variables.
- Checked for missing values.
- Confirmed the absence of duplicate records.
- Investigated potential outliers and extreme ticket prices.
- Performed feature engineering by categorizing departure times into interpretable time periods.
- Generated additional categorical variables for flight duration and ticket price segmentation.

The dataset was subsequently validated and found suitable for exploratory analysis.

Exploratory Analysis and Key Findings

1. Airline Market Distribution

The analysis began by examining the frequency distribution of airlines operating within the dataset.

Findings

- The dataset contains multiple airlines with varying market presence.
- A few airlines account for a substantially larger proportion of total flights.
- Premium and budget airlines coexist within the market, creating considerable variation in ticket pricing.

Business Insight

Higher operational frequency does not necessarily correspond to higher ticket prices. Airlines adopt distinct pricing strategies depending on their target customer segments.

2. Average Ticket Price by Airline

Average airfare was calculated for every airline to identify pricing differences.

Findings

- **Vistara recorded the highest average ticket price** among all airlines.
- Premium carriers consistently charged significantly higher fares than low-cost airlines.
- Budget airlines maintained comparatively lower average ticket prices.

Business Interpretation

The results indicate that airline brand positioning and service quality play a substantial role in airfare determination. Premium airlines appear to compete primarily through enhanced customer experience rather than price.

3. Flight Distribution by Source City

The project analyzed the number of flights originating from each source city.

Findings

- Major metropolitan cities contribute the largest share of scheduled flights.
- Flight availability is concentrated around high-demand transportation hubs.

Business Insight

Cities with higher passenger demand naturally exhibit greater airline activity, reflecting the concentration of commercial aviation networks.

4. Effect of Number of Stops on Ticket Price

The relationship between flight connectivity and airfare was investigated through average price comparisons and boxplot analysis.

Findings

- Ticket prices generally increase with the number of stops.
- Flights involving multiple stops exhibit greater variability in prices.
- Non-stop flights are not always the most expensive; however, connecting flights often command higher average fares.

Business Interpretation

Additional stops frequently correspond to longer travel distances, more complex routing, and increased operational costs, contributing to higher ticket prices.

5. Effect of Flight Duration on Ticket Price

The relationship between flight duration and airfare was explored using scatter plots, regression analysis, duration categorization, boxplots, and correlation analysis.

Findings

- Longer flights generally exhibit higher ticket prices.
- Scatter plots reveal an upward pricing trend as duration increases.
- Regression analysis confirms a positive relationship between duration and airfare.
- Correlation analysis indicates that the relationship is **positive but relatively weak**, suggesting that flight duration alone does not fully explain ticket pricing.
- Longer duration categories tend to have higher median ticket prices.

Business Interpretation

Although flight duration contributes to airfare variation, airline pricing depends on multiple factors beyond travel time, including airline positioning, route demand, competition, and service class.

6. Departure Time Analysis

Departure times were categorized into different periods of the day to evaluate their influence on pricing.

Findings

- Average ticket prices differ across departure periods.
- Certain departure windows consistently exhibit higher average fares.
- These differences likely reflect variations in passenger demand and scheduling preferences.

Business Interpretation

Flights operating during high-demand time periods can command premium prices because they better align with passenger convenience.

7. Distribution of Ticket Prices

Ticket prices were examined using histograms, kernel density estimation (KDE), boxplots, and skewness analysis.

Findings

- Ticket prices display a **positively skewed distribution**.
- Most tickets are concentrated within a moderate price range.
- A relatively small number of premium-priced flights create a long right tail.
- Boxplot analysis confirms the presence of high-value observations.

Business Interpretation

The airline market is characterized by substantial pricing heterogeneity, where premium fares significantly increase overall price variability.

8. Factors Contributing to Expensive Tickets

To identify drivers of expensive tickets, flights were classified into higher-priced and lower-priced categories using the median ticket price.

Cross-tabulation and proportional analyses were then performed.

Findings

- **Vistara contributes the highest proportion of expensive tickets**, reinforcing its premium market positioning.
- **Air India** also records a comparatively large share of higher-priced flights.
- Budget airlines contribute a much larger proportion of lower-priced tickets.

- Flights involving multiple stops are more frequently associated with expensive fare categories.

Business Interpretation

Premium pricing is influenced primarily by airline positioning, while flight connectivity further contributes to fare escalation.

9. Correlation Analysis

A correlation heatmap was generated to evaluate relationships among numerical variables.

Findings

- Flight duration exhibits a weak positive relationship with ticket price.
- Other numerical variables display relatively low correlations.
- No evidence of severe multicollinearity was observed.

Business Insight

Ticket prices are influenced by a combination of interacting operational and market variables rather than any single numerical factor.

Overall Business Insights

The exploratory analysis successfully identified several important determinants of airline ticket pricing.

Major Insights

- **Airline selection is the strongest determinant of airfare**, with premium carriers consistently charging higher prices.
- **Longer-duration flights generally cost more**, although duration alone explains only part of the observed price variation.
- **Flights with additional stops tend to have higher average ticket prices** and greater price variability.
- **Departure period influences pricing**, indicating the effect of passenger demand and scheduling preferences.
- **Ticket prices exhibit significant positive skewness**, reflecting the presence of premium-priced flights.
- **Major metropolitan cities account for the highest flight volumes**, emphasizing the concentration of airline operations around key transportation hubs.

- **Premium airlines such as Vistara and Air India contribute disproportionately to expensive ticket categories**, highlighting the importance of market positioning in pricing strategy.
 - **Correlation analysis demonstrates that airfare is driven by multiple interacting factors**, underscoring the complexity of airline pricing.
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Conclusion

This project demonstrates how systematic exploratory data analysis can transform large-scale aviation data into actionable business intelligence. By **cleaning, validating, engineering, visualizing, investigating, quantifying, segmenting, comparing, and interpreting** over **300,000 flight records**, the analysis uncovered meaningful pricing patterns that can support airline pricing strategy, demand forecasting, route planning, and customer decision-making.

The project also showcases practical proficiency in **Python, Pandas, NumPy, Matplotlib, Seaborn, feature engineering, statistical analysis, correlation analysis, and business-focused data storytelling**, while establishing a strong foundation for future work in **predictive fare modeling, machine learning, and revenue optimization**.

Data Attributes and Verification

The exploratory analysis is based on a domestic aviation dataset encompassing several critical operational and market features:

- Airline carrier identification
- Origin and destination metropolitan areas
- Scheduled departure and arrival periods
- Flight connectivity (number of stops)
- Total travel duration
- Booking lead time (days remaining)
- Observed ticket pricing

To maintain analytical integrity, the raw data was systematically audited for:

- Presence of null or missing values
- Redundant or duplicated flight records
- Consistency and correctness of data types
- Identification of statistical outliers and anomalies

Standardized cleaning protocols were implemented to ensure the reliability of the subsequent findings.

Comprehensive Exploratory Analysis

1. Operational Frequency by Airline

The initial investigation quantified the market presence of various carriers within the study period.

Business Insight

- Market concentration is high, with specific airlines accounting for the majority of flight traffic.
- Smaller carriers maintain a limited footprint within the current dataset.

Analytical Visualization: *Categorical Bar Distribution*

2. Airfare Variation Across Carriers

Average fare metrics were established for each airline to detect pricing disparities.

Business Insight

- **Vistara** consistently maintains a premium pricing position relative to the market.
- Low-cost carriers successfully differentiate through significantly lower average price points.

Analytical Visualization: *Mean Price Comparison Chart*

3. Spatial Distribution of Flight Volume

The analysis mapped the concentration of flight originations across various hub locations.

Business Insight

- Flight density is heavily skewed toward major metropolitan transportation hubs.
- Airline activity aligns with the passenger demand profiles of high-density urban centers.

Analytical Visualization: *Geographic Frequency Count Plot*

4. Relationship Between Connectivity and Fare

Airfare sensitivity was evaluated against varying levels of flight stops, including direct and connecting services.

Findings

- A positive correlation exists between the number of stops and overall ticket cost.
- Median fares escalate as routing complexity increases.
- Connecting flights exhibit significantly higher price volatility compared to non-stop options.

Analytical Visualization: *Interquartile Range Boxplot Analysis*

5. Correlation of Travel Time and Price

The association between journey duration and fare was quantified through statistical and visual modeling.

Findings

- Extended flight durations are generally associated with higher pricing tiers.
- Regression analysis indicates a positive, albeit modest, relationship.
- Pearson coefficients suggest that duration is only a partial driver of fare variation.

Analytical Visualizations: *Scatter Plots and Linear Regression Models*

6. Temporal Departure Analysis

Flights were segmented into daily time intervals (e.g., Morning, Evening, Night) to assess temporal demand influence.

Findings

Distinct pricing trends emerge across different departure windows, reflecting peak demand periods and passenger scheduling utility.

Analytical Visualization: *Temporal Mean Fare Distribution*

7. Statistical Distribution of Fares

The project analyzed the overall price structure using density estimation and skewness metrics.

Findings

- Airfare data demonstrates **substantial positive skewness**.
- The majority of ticket transactions are clustered at lower to moderate price points.
- A long right tail indicates a subset of high-value, premium-priced flights.

Analytical Visualizations: *Histogram and Kernel Density Estimation (KDE)*

8. Determinants of Premium Ticket Pricing

Comparative analysis was conducted across airlines and connectivity levels to isolate the drivers of high-cost fares.

Findings

- Premium-positioned airlines are the primary contributors to the high-price segment.
- Indirect routing and multiple stops further correlate with increased ticket costs.
- Airline brand remains a more significant price determinant than simple flight duration.

Analytical Visualizations: *Proportional Stacked Charts and Cross-Tabulation*

9. Feature Correlation Matrix

A systematic evaluation of the interrelationships between numerical variables was performed.

Key Observations

- Duration maintains a positive, though relatively weak, association with price.
- Lead time (days remaining) serves as a potent predictor of fare variation.
- There is no evidence of excessive collinearity within the feature set.

Analytical Visualization: *Correlation Matrix Heatmap*

Major Business Findings

The systematic EDA revealed several critical drivers of airline pricing strategy:

- **Airline brand positioning** is the primary determinant of airfare levels.
- **Route duration** typically scales with price, although other operational factors exert significant influence.
- **Connectivity complexity** (stops) is positively associated with higher ticket costs.

- **Price skewness** is notable, with premium segments driving overall variance.
 - **Strategic hub concentration** is evident, as metropolitan cities dominate total volume.
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Analytical Expertise Applied

This initiative utilizes advanced data science proficiencies including:

- Data quality auditing and cleaning
 - Exploratory Data Analysis (EDA) methodologies
 - Advanced feature engineering and binning
 - Multi-variate statistical interpretation
 - Quantitative data storytelling
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Technology Stack

- Python Core Programming
- Pandas and NumPy (Data Manipulation)
- Seaborn and Matplotlib (Data Visualization)
- Jupyter Notebook (Interactive Development)